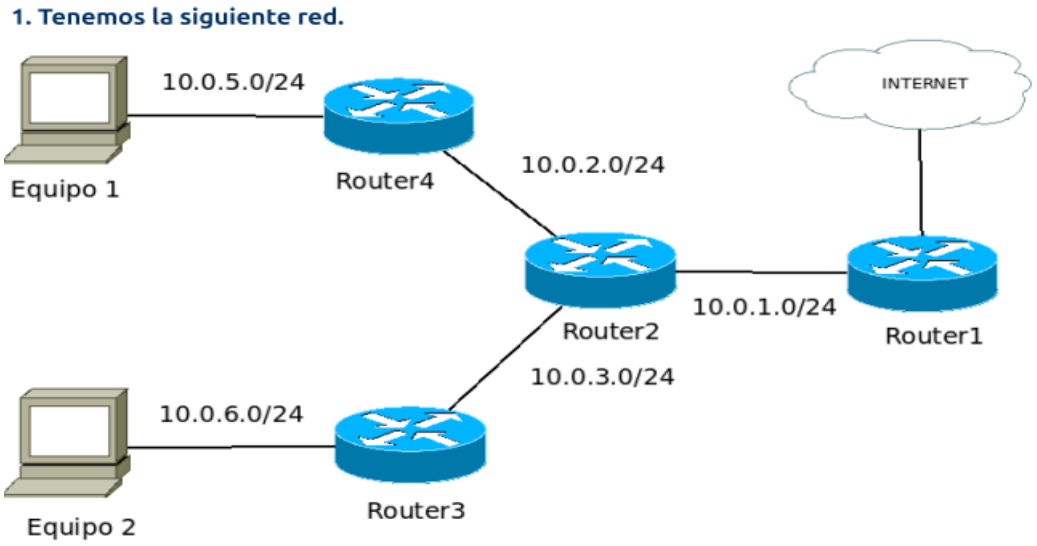
Esquema:



Configuración física

1. configuración física router1:

```
Red
Adaptador 1:   Intel PRO/1000 MT Desktop (Adaptador solo anfitrión, «VirtualBox Host-Only Ethernet Adapter»)
Adaptador 2:   Intel PRO/1000 MT Desktop (NAT)
Adaptador 3:   Intel PRO/1000 MT Desktop (Red interna, «red1»)
Adaptador 4:   Intel PRO/1000 MT Desktop (No conectado)
```

   En clase el adaptador 2 en vez de NAT será un adaptador puente para que salga a internet por el PC del profesor.

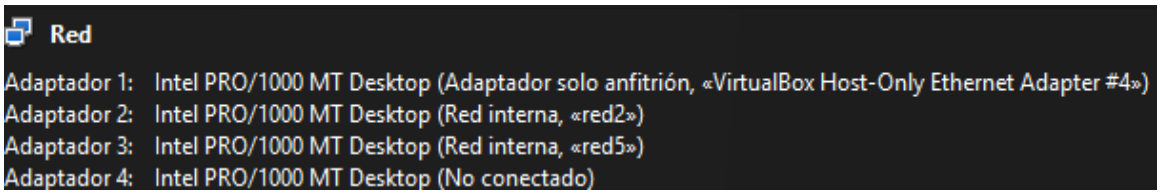
2. configuración física router2:

```
Red
Adaptador 1:   Intel PRO/1000 MT Desktop (Adaptador solo anfitrión, «VirtualBox Host-Only Ethernet Adapter #2»)
Adaptador 2:   Intel PRO/1000 MT Desktop (Red interna, «red1»)
Adaptador 3:   Intel PRO/1000 MT Desktop (Red interna, «red2»)
Adaptador 4:   Intel PRO/1000 MT Desktop (Red interna, «red3»)
```

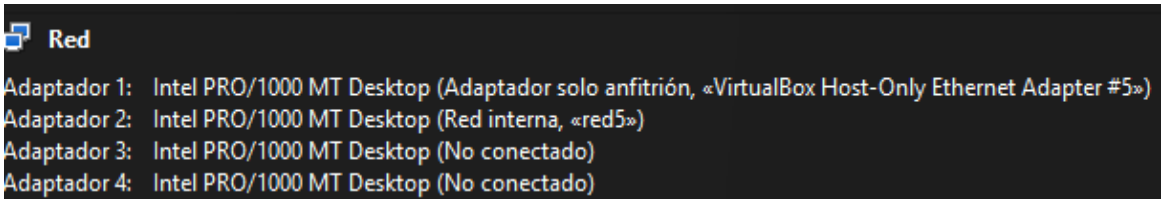
3. configuración física router3:

```
Red
Adaptador 1:   Intel PRO/1000 MT Desktop (Adaptador solo anfitrión, «VirtualBox Host-Only Ethernet Adapter #3»)
Adaptador 2:   Intel PRO/1000 MT Desktop (Red interna, «red3»)
Adaptador 3:   Intel PRO/1000 MT Desktop (Red interna, «red4»)
Adaptador 4:   Intel PRO/1000 MT Desktop (No conectado)
```

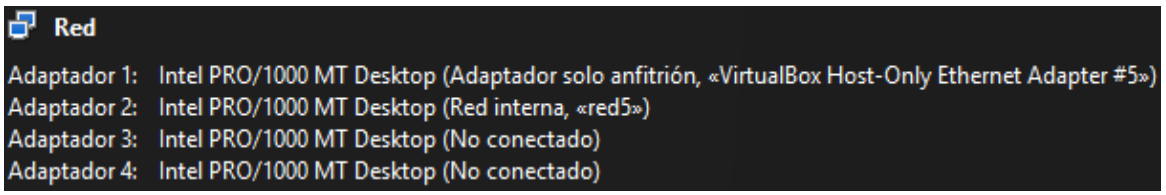
4. configuración física router4:



5. configuración física equipo1:



6. configuración física equipo2:



Configuración lógica

1. Cambiamos el nombre del router o dispositivo (/etc/hosts y /etc/hostname).

```
 GNU nano 7.2                                                    /etc/hosts
127.0.0.1 localhost
127.0.1.1 Router1

# The following lines are desirable for IPv6 capable hosts
::1     ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

```
 GNU nano 7.2                                                    /etc/hostname
Router1
```

2. procedemos a configurar las interfaces (/etc/netplan/50-cloud-init.yaml).
   - Router1

```
network:
    ethernets:
        enp0s3:
            dhcp4: true
        enp0s8:
            dhcp4: true
        enp0s9:
            dhcp4: false
            addresses:
              - 10.0.1.1/24
            nameservers:
              addresses: [1.1.1.1, 8.8.8.8]
            routes:
              - to: 10.0.2.0/24
                via: 10.0.1.2
              - to: 10.0.3.0/24
                via: 10.0.1.2
              - to: 10.0.5.0/24
                via: 10.0.1.2
              - to: 10.0.6.0/24
                via: 10.0.1.2
        enp0s10:
            dhcp4: true
    version: 2
```

   - Router2

```
network:
    ethernets:
        enp0s3:
            dhcp4: true
        enp0s8:
            dhcp4: false
            addresses:
              - 10.0.1.2/24
            nameservers:
              addresses: [1.1.1.1, 8.8.8.8]
            routes:
              - to: default
                via: 10.0.1.1
        enp0s9:
            dhcp4: false
            addresses:
              - 10.0.2.1/24
            nameservers:
              addresses: [1.1.1.1, 8.8.8.8]
            routes:
              - to: 10.0.5.0/24
                via: 10.0.2.2
        enp0s10:
            dhcp4: false
            addresses:
              - 10.0.3.1/24
            nameservers:
              addresses: [1.1.1.1, 8.8.8.8]
            routes:
              - to: 10.0.6.0/24
                via: 10.0.3.2
    version: 2
```

- Router3

```
network:
    ethernets:
        enp0s3:
            dhcp4: true
        enp0s8:
            dhcp4: false
            addresses:
              - 10.0.3.2/24
            nameservers:
              addresses: [1.1.1.1, 8.8.8.8]
            routes:
              - to: default
                via: 10.0.3.1
        enp0s9:
            dhcp4: false
            addresses:
              - 10.0.6.1/24
        enp0s10:
            dhcp4: true
    version: 2
```

- Router4

```
network:
    ethernets:
        enp0s3:
            dhcp4: true
        enp0s8:
            dhcp4: false
            addresses:
              - 10.0.2.2/24
            nameservers:
              addresses: [1.1.1.1, 8.8.8.8]
            routes:
              - to: default
                via: 10.0.2.1
        enp0s9:
            dhcp4: false
            addresses:
              - 10.0.5.1/24
        enp0s10:
            dhcp4: true
    version: 2
```

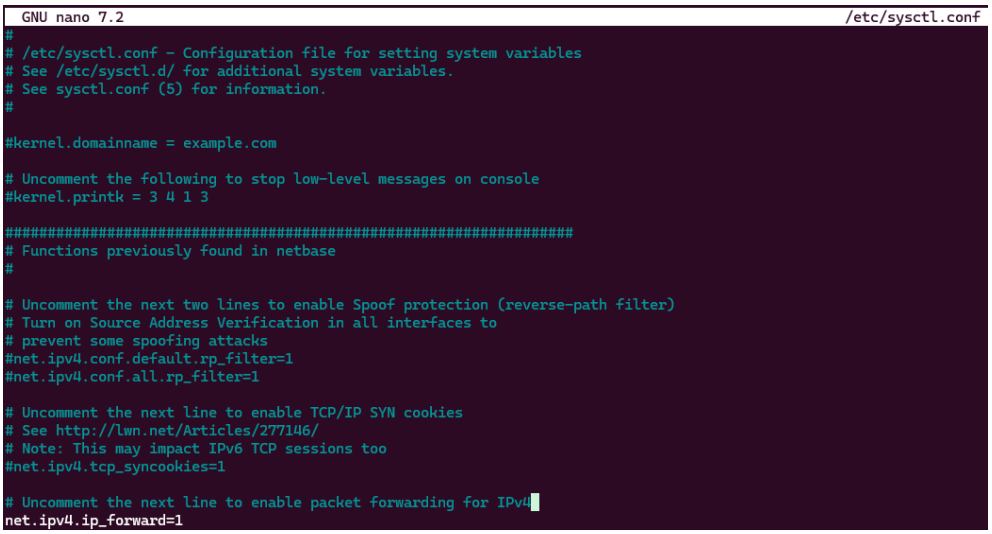
- Equipo1

```
network:
    ethernets:
        enp0s3:
            dhcp4: true
        enp0s8:
            dhcp4: false
            addresses:
              - 10.0.5.2/24
            nameservers:
              addresses: [1.1.1.1, 8.8.8.8]
            routes:
              - to: default
                via: 10.0.5.1
        enp0s9:
            dhcp4: true
        enp0s10:
            dhcp4: true
    version: 2
```

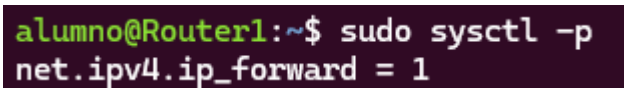
- Equipo2

```
network:
    ethernets:
        enp0s3:
            dhcp4: true
        enp0s8:
            dhcp4: false
            addresses:
              - 10.0.6.2/24
            nameservers:
              addresses: [1.1.1.1, 8.8.8.8]
            routes:
              - to: default
                via: 10.0.6.1
        enp0s9:
            dhcp4: true
        enp0s10:
            dhcp4: true
    version: 2
```

3. Para que haya comunicación entre los routers hay que activar el reenvío de paquetes:
    - sudo nano ***/etc/sysctl.conf*** (la opción estará comentada"#")



    - para ver que esta activado y funcionando ejecutamos ***(sudo sysctl -p)***



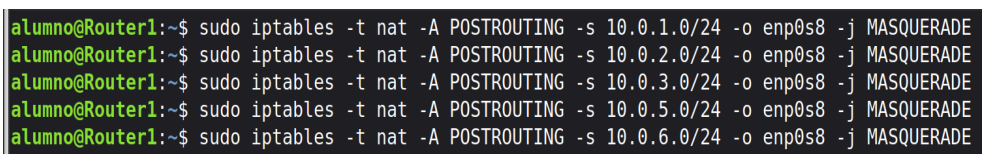
4. Para que el router1 saque los paquetes a internet de otros routers que no sea el mismo hay que activar el enmascaramiento:

    - ***sudo iptables -t nat -A POSTROUTING -o enp0s8 -j MASQUERADE***

    - Hay que tener en cuenta que la interfaz que debes poner es la que sale a internet, en este caso es la enp0s8.

    - Lo use tanto en el R1 y R2

    - ***sudo iptables -t nat -A POSTROUTING -s 10.0.1.0/4 -o (interfaz) -j MASQUERADE***
      [así con todas las redes]



- Para guardar esta configuración de iptables:
  se descarga el paquete:

  Instaralacion: ***apt-get install iptables-persistent***
  y se guardará en el la ruta:
  /etc/iptables/rules.v4

```
alumno@Router1:~$ sudo cat  /etc/iptables/rules.v4
# Generated by iptables-save v1.8.10 (nf_tables) on Tue Sep 29 10:01:42 2026
*nat
:PREROUTING ACCEPT [0:0]
:INPUT ACCEPT [0:0]
:OUTPUT ACCEPT [0:0]
:POSTROUTING ACCEPT [16:1028]
-A POSTROUTING -s 10.0.1.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.2.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.3.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.5.0/24 -o enp0s8 -j MASQUERADE
-A POSTROUTING -s 10.0.6.0/24 -o enp0s8 -j MASQUERADE
COMMIT
# Completed on Tue Sep 29 10:01:42 2026
```

  para guardar los postrouting se hace:
  sudo netfilter-persistent save

```
alumno@Router1:~$ sudo netfilter-persistent save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/15-ip4tables save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/25-ip6tables save
```

  Para ver las tablas de reglas tenemos este comando:
  sudo iptables -t nat -L -nv

```
alumno@Router1:~$ sudo iptables -t nat -L -nv
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain POSTROUTING (policy ACCEPT 20 packets, 2064 bytes)
 pkts bytes target     prot opt in     out     source               destination
    8   604 MASQUERADE  0    --  *      enp0s8  10.0.1.0/24          0.0.0.0/0
    7   528 MASQUERADE  0    --  *      enp0s8  10.0.2.0/24          0.0.0.0/0
    8   584 MASQUERADE  0    --  *      enp0s8  10.0.3.0/24          0.0.0.0/0
    7   528 MASQUERADE  0    --  *      enp0s8  10.0.5.0/24          0.0.0.0/0
    8   584 MASQUERADE  0    --  *      enp0s8  10.0.6.0/24          0.0.0.0/0
```